

Surya Prakash Kahar

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Professional Summary

Results-driven Computer Science undergraduate at IIIT Bhagalpur (**CGPA: 8.0**) specializing in **Distributed Systems** and **Cloud-Native Backend Architectures**. **LeetCode Knight (1704)** with expertise in **Go (Golang)**, **C++**, and **MERN Stack**. Proven track record in contributing to high-traffic open-source ecosystems (**850k+ users**) and engineering fault-tolerant microservices. Expert in **Raft Consensus**, **ACID-compliant databases**, and **real-time AI-integrated pipelines**.

Education

Indian Institute of Information Technology (IIIT) Bhagalpur

Bhagalpur, Bihar

Bachelor of Technology in Computer Science and Engineering (CGPA: 8.0/10.0)

July 2023 – July 2027

Technical Skills

Languages: C, C++, Go (Golang), SQL (PostgreSQL, MySQL), JavaScript (ES6+), Python, HTML/CSS

Backend & Systems: Node.js, Express.js, Raft Consensus, gRPC, REST APIs, JWT, Redis, ACID, Socket.io

Infrastructure & DevOps: Docker, Docker Compose, AWS (EC2, S3), Git/GitHub, CI/CD Pipelines, Linux/Bash

Frontend & Tools: React.js, Redux, ImageKit CDN, Postman, Vercel, Framer Motion, Chrome Web Store API

Core CS: Data Structures & Algorithms (DSA), System Design, OS, DBMS, Computer Networks, OOP

Experience

Software Development Engineer Intern (Recreivo)

Bengaluru

Full-Stack Engineering

April 2026 – Present

- Developed and integrated RESTful APIs using Node.js and Express.js, improving data retrieval efficiency by 20% for core application services.
- Collaborated with the frontend team to implement responsive UI components using React.js, enhancing overall user experience and component reusability.
- Participated in Agile sprint planning and conducted peer code reviews to ensure maintainability, security, and adherence to software engineering best practices.

Open Source Contributor (CodeCharity)

Remote

ImprovedTube – YouTube Browser Extension | JavaScript, Web Extensions, Git

2025 – Present

- Optimized client-side performance for a global browser extension serving **850,000+ active users**, reducing memory overhead in content script execution.
- Engineered a context-aware **Smart Speed** engine utilizing video metadata analysis to implement adaptive playback control, enhancing user engagement and retention.
- Resolved complex UI regressions and deep-link inconsistencies in **YouTube Shorts/Playlists**, ensuring robust cross-platform stability across Reddit and external website embeds.
- Collaborated via **GitHub Agile workflows**, performing technical code reviews and patching security vulnerabilities to maintain a high quality of production code.

Technical Projects

Distributed Raft Consensus Engine | Go (Golang), TCP/RPC, BoltDB, Docker, React.js

2026

- Architected a **fault-tolerant** distributed Key-Value store in Go, implementing the Raft Algorithm to ensure strict data consistency and 99.9% availability across a multi-node cluster.
- Developed an asynchronous networking layer over **TCP/RPC**, handling leader election, log replication, and automated safety invariants to prevent split-brain/stale-leader scenarios.
- Built a real-time **React.js** visualizer using telemetry data to map state transitions and RPC heartbeats.

Scalable Bank Transaction Engine | Node.js, PostgreSQL, Redis, JWT, Middleware

2026

- Engineered a secure financial backend using **PostgreSQL transactions** for ACID compliance, eliminating race conditions in concurrent fund transfers for 1,000+ simulated users.
- Optimized system throughput by 30% using strategic indexing and implemented **Redis-based rate limiting** to safeguard against brute-force and DDoS attack vectors.

Certifications & Training

- AI and Machine Learning Algorithms** (Microsoft): Specialized in Neural Networks and Predictive Analytics.
- Meta Full Stack Professional Certificate** (Meta): End-to-End Database Management and RESTful API Design.
- Generative AI for Expert Partners** (Google): Expertise in LLM deployment, Prompt Engineering, and Vertex AI.

Honors and Achievements

- Competitive Programming:** Max Rating **1704 (Knight)** on LeetCode with 550+ problems solved; **3-Star Coder** on CodeChef; Global Rank top 5% in multiple coding contests.